ATLAS Note

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Centrality Determination in $\sqrt{s_{\mathrm{NN}}} = 5.36$ TeV Pb+Pb Collisions from 2023 data running

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Supporting note for the determination of centrality and the geometric quantities (N_{part}, N_{coll}, T_{AA}) in 2023 data running.

Contents

1 Introduction

This note describes the geometric model for 5.32 TeV Pb-Pb collision and its application to the data collected by the ATLAS experiment in 2023. The centrality of the collisions is characterized by the sum of the transverse energy in both modules of the Forward Calorimeter, situated at $3.2 < |\eta| < 4.9$. This largely follows the procedure used in Run2 and Run1. As previously, the different detector performance, running conditions, larger $\sqrt{s}$ and different minimum bias trigger conditions require a dedicated new analysis. The Glauber simulation is matched in the region ΣE_T >30 GeV which is estimated to contain 89% ± 1% of the total Gluaber-like cross section. The ΣE_T values corresponding to centile cuts, as well as the values of N_{coll}, N_{part} and T_{AA} in different centrality bins are presented, along with the corresponding systematic uncertainties.

2 Data Selection

The FCal ΣE_T distribution was constructed by selecting data from the PC and CC streams using the nominal MinBias triggers. For the 2023 datataking those were:

- HLT_mb_sptrk_pc_L1ZDC_A_C_VTE50
- HLT_noalg_L1TE600p0ETA49
- HLT_noalg_L1TE50_VTE600p0ETA49

The containers used were the :

- data23_hi.periodAllYear2.physics_PC.PhysCont.AOD.repro35_v01
- data23_hi.periodAllYear2.physics_CC.PhysCont.AOD.repro35_v01

The data is then influenced primarily by in-time and out-of-time pileup effects. We proceed with removing the in-time pileup which can be done by cutting on the ZDC-FCal ΣE_T distribution.

2.1 In-time pileup removal

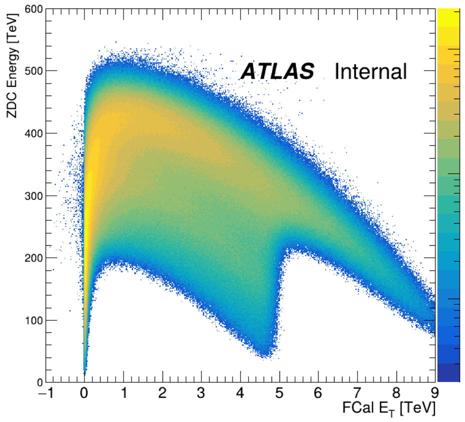
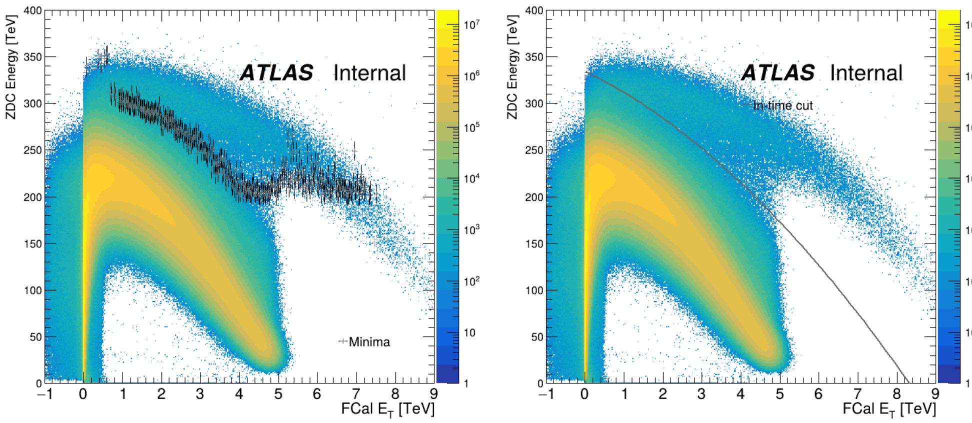


Figure 1: An approximation of the ZDC-FCal ΣE_T distribution of purely in-time pileup events.



(a) ZDC-FCal ΣE_T distribution with minima of the distributions overlaid.

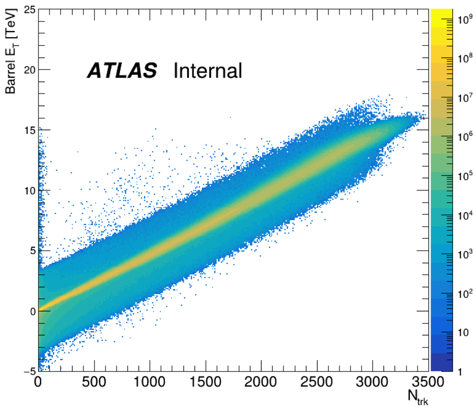
(b) ZDC-FCal ΣE_T with the fitted curve overlaid.

Figure 2: The left figure shows the ZDC-FCal ΣE_T distribution with the minima overlaid. The right figure shows the ZDC-FCal ΣE_T distribution with the resultant fitted curve overlaid.

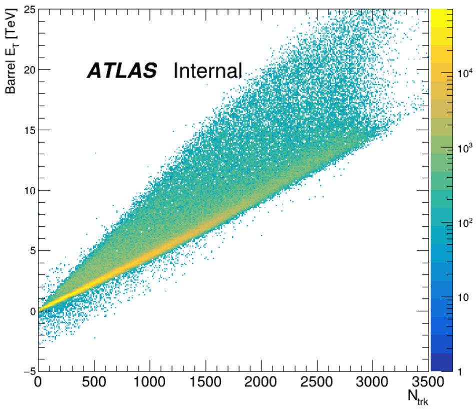
To begin, the in-time pileup distribution must be approximated. The ZDC-FCal ΣE_T distribution is convoluted with itself to simulate a second event being overlaid on the first, shown in Fig. 1. An examination of the ZDC-FCal ΣE_T distribution shows that there exists a minimum between the two arms of the distribution, which can be isolated and used to cut the data. This is shown in Fig. 2(a). This minimum distribution can be fit with a polynomial:

$$f(x) = a + b \cdot x + c \cdot x^2 \tag{1}$$

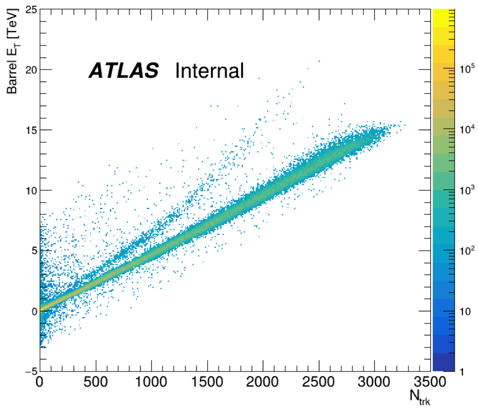
with a=334.29 TeV, b=-20.39, c=-2.38 TeV^{-1}. This fit can be then used to cut in-time pileup by requiring ZDC Energy < f(FCal ΣE_T). The effects of this cut on the distribution of Barrel E_T against N_{trk} is shown in Fig. 3(b). We can see the cut has removed a significant portion of events falling outside of the main band of the distribution, as well as some events within it. The distribution post cut still contains some events outside of the main bands which are suppressed by 10^{-4}, so we can safely neglect them.



(a) N_{trk} against Barrel ΣE_T after cut.



(b) N_{trk} against Barrel ΣE_T of events cut due to in-time pileup cut.



(c) N_{trk} against Barrel ΣE_T of events due to out-of-time pileup cut.

Figure 3: The effects of the in-time pileup cut on the Barrel E_T distribution.

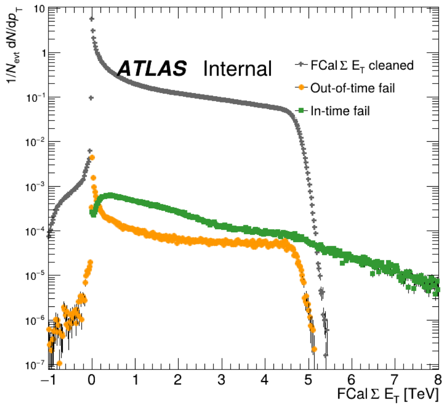


Figure 4: The final FCal ΣE_T distribution after both in-time and out-of-time pileup cuts.

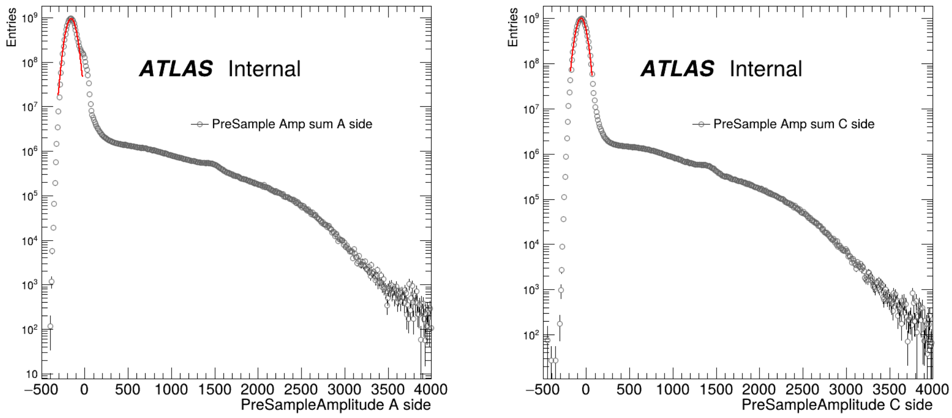
2.2 Out-of-time pileup removal

Out-of-time removal requires a more careful study. The ZDC PreSampleAmp distribution was used here, shown in Fig. 5. The distribution can be obtained by summing the presample amplitudes from all four ZDC modules in each calorimeter. The xAOD variable which contains this information is the **PreSampleAmp**, which can be obtained from the **ZdcModule** container. This distribution shows one peak which correspond to in-time collisions and a long tail correspoinding to positive energy out-of-time pileup events, from previous bunch crossings. This long tail shows additional electromagnetic energy deposited in the ZDC, which is a signature of an out-of-time pileup event. This is due to the fact that the rate of electromagnetic pileup in a single ZDC is much higher than the hadronic pileup rate, due to the large cross section for single dissociation. The distribution on both sides of the ZDC is first fit with a Gaussian in the initial peak. Then, everything more than 5 σ away from the peak is cut. This is shown in Fig. 6. The effects of this cut on the Barrel ΣE_T vs N_{trk} distribution are shown in Fig. 3(c). The results of both in-time and out-of-time cuts are shown in Fig. 4. The final distribution can be used for the centrality analysis.

3 Glauber Model

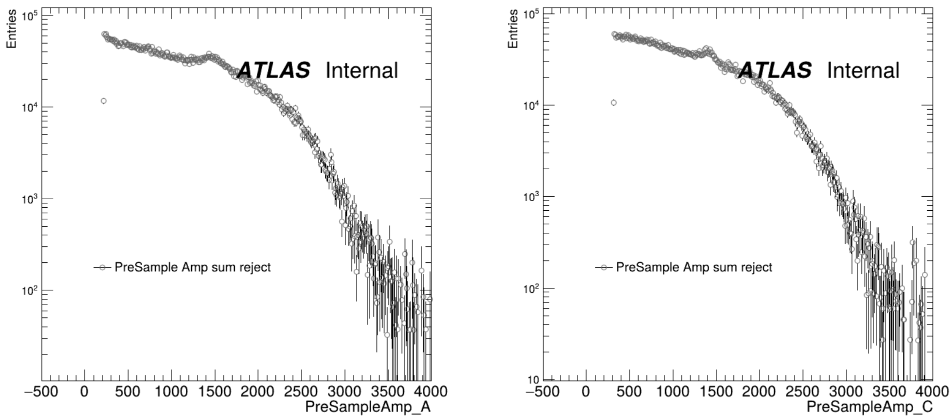
One million events of Pb+Pb collisions were simulated at 5.36 TeV energy using the PHOBOS GlauberMC v 3.2 code[1], with a nucleon-nucleon cross section of $\sigma_{NN} = 68.1$, Woods-Saxon parameters $(R, a) =$ (6.62 fm, 0.546 fm) and nucleon hard core $d_N = 0.4$ fm[2]. Events are characterized by the number of participating nucleons N_{part} and the number of binary nucleon-nucleon collisions N_{coll}. Fig. 7(a) and 7(b) shows the distributions over N_{part} and N_{coll}. The so-called Two-Component model (TCM) is used to map the distributions of the geometric parameters onto the distributions from the data. In brief, the TCM postulates that the rate of soft particle production scales linearly with a hybrid geometric quantity which is a linear combination of N_{part} and N_{coll}. This quantity, called the number of fundamental elements of particle production is written

$$N_{\text{elem}} = xN_{\text{coll}} + \frac{1}{2}(1-x)N_{\text{part}} \tag{2}$$



(a) (b)

Figure 5: The ZDC PreSampleAmp distribution for the A (left) and C (right) sides.



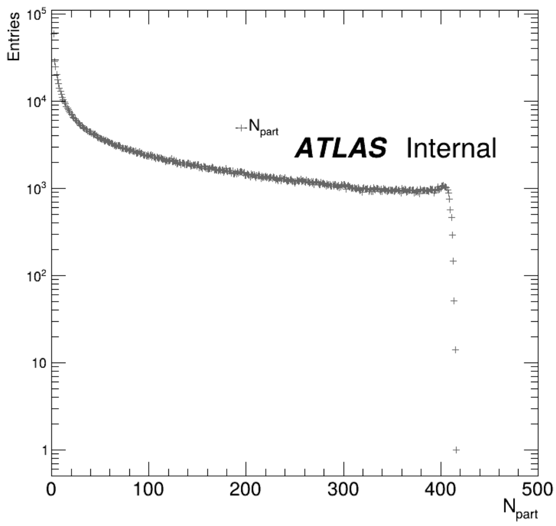
(a) (b)

Figure 6: The ZDC PreSampleAmp distribution for the A (left) and C (right) sides after the cut.

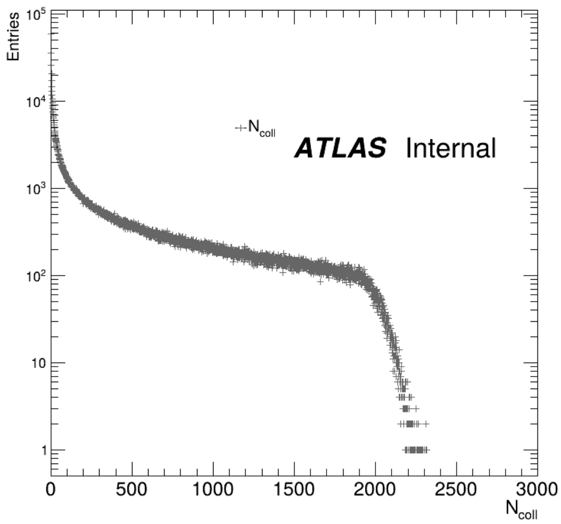
where the variable x controls the linear weighting. 7(c) shows the resultant N_{elem} distribution. For pp collisions, where $N_{\mathrm{part}} = 2$ and $N_{\mathrm{coll}} = 1$, $N_{\mathrm{elem}} = 1$ for all values of x. Thus, the ΣE_T produced per element is equivalent to that in a single pp collision at the same $\sqrt{s}$. The ΣE_T distribution from a single element of particle production is modeled in two ways, either as a Gaussian or a di-Gamma distribution.

$$P(\Sigma E_T; N_{\mathrm{elem}} = 1) = \mathrm{Gaussian}(\Sigma E_T; \sigma; \mu) \tag{3}$$

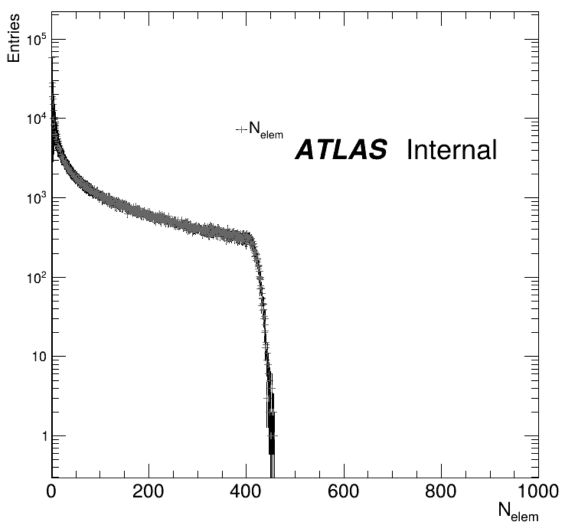
$$P(\Sigma E_T; N_{\mathrm{elem}} = 1) = \mathrm{Gamma}(\Sigma E_T; \Theta; k) \tag{4}$$



(a)



(b)



(c)

Figure 7: Distributions of the geometric quantities $N_{\rm part}$ and $N_{\rm coll}$ in Pb+Pb collisions at 5.36 TeV. The $N_{\rm elem}$ distribution is shown below.

Each element is treated as producing particles independently. Thus, the ΣE_T distribution from an event with $N_{\text{elem}} > 1$ is an N_{elem}-fold convolution over the original distribution.

$$P(\Sigma E_T; N_{\text{elem}}) = \text{Gaussian}(\Sigma E_T; \sigma\sqrt{N_{\text{elem}}}; \mu N_{\text{elem}}) \tag{5}$$

$$P(\Sigma E_T; N_{\text{elem}}) = \text{Gamma}(\Sigma E_T; \Theta; kN_{\text{elem}}) \tag{6}$$

The ΣE_T distribution over all Pb+Pb events is given by the linear combination of the ΣE_T distribution from each N_{elem} value, weighted by the probability of that value of N_{elem} in Pb+Pb events.

$$P(\Sigma E_T) = \sum_{N_{\text{elem}}} P(\Sigma E_T; N_{\text{elem}}) P(N_{\text{elem}}) \tag{7}$$

3.1 Fit parameters

The parameters of the response model are obviously an important part of the analysis. The fitting procedure is done by a χ^2 minimization of the smeared N_{elem} distribution with the data. The fit finds the best values to be

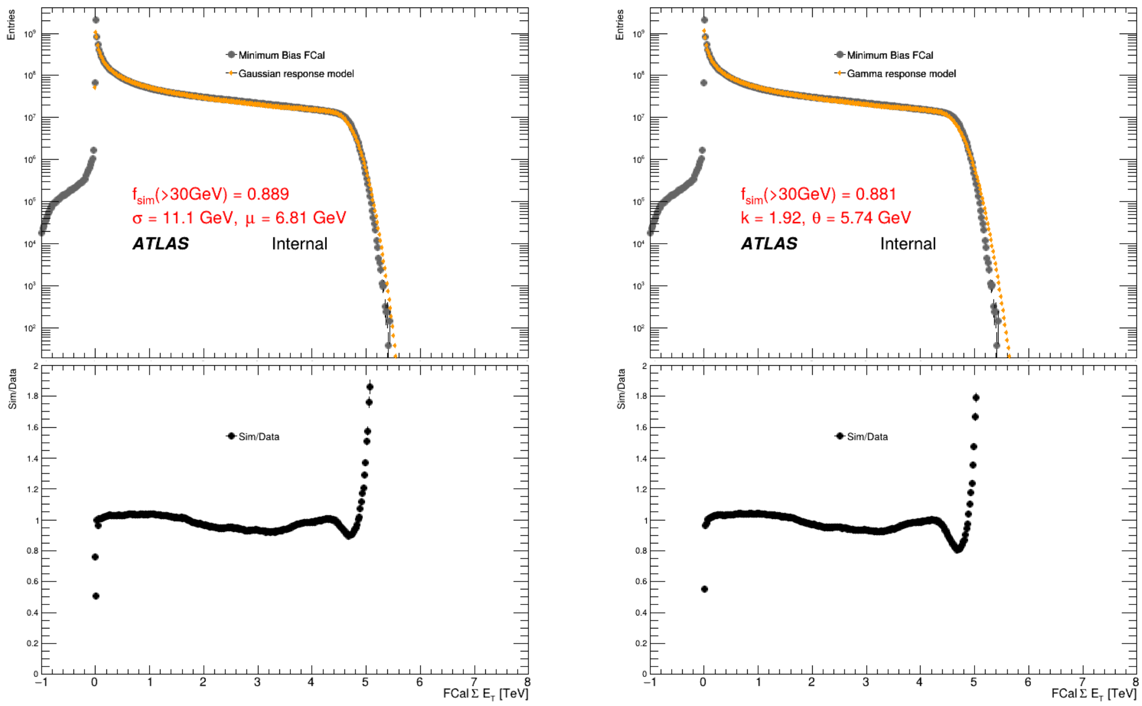
- Gaussian: σ = 11.1 GeV, μ = 5.9 GeV
- Di-Gamma: k = 1.95 , θ = 5.7 GeV

3.2 Choice of x

The choice of x is a major factor in determining the final shape of the N_{elem} distribution. Its determination can be in two ways: by relating the $\langle \Sigma E_T \rangle_{pp}$ and $\langle \Sigma E_T \rangle_{PbPb}$, or by varying the x in finite steps and allowing the Gaussian/Di-Gamma parameters to float while keeping the shoulder of the N_{elem} as flat as possible[3]. The former method is more robust, but requires a measurement of $\langle \Sigma E_T \rangle_{pp}$ which is not biased by the choice of triggers, which is nearly impossible. The second, although less robust, is more practical. The value of x found in this analysis is 0.085 ± 0.005. Since the previous analysis used the value of 0.09, we set this as the value.

3.3 Results

The results of the fit are shown in Fig. 8. The Gaussian response model was chosen, as it more closely matches the data distribution in the shoulder region. The tool used to create this analysis is the CentralityFittingTool[4].



(a) (b)

Figure 8: Fit results showing a Glauber + Gaussian response model, Fig. 8(a) and Glauber + Di-Gamma response model, Fig 8(b). The ratio of the data to the fit is shown below.

4 Centrality Cuts

The ΣE_T values which correspond to the centile positions were determined from the data. The minimum bias Pb+Pb events above the working point were sorted by their ΣE_T values. Since the nominal fit finds an efficiency of 89%, the events were divided into 89 bins of equal sizes, and the dividing values were taken to be the positions of the centile cut. In order to allow for future refinements of the efficiency, we define the centrality up to 85% at this time. The centrality cuts are shown in Table 1.

1% intervals	
Centile cut	FCalΣE_T
85%	0.0388482
84%	0.0430012
83%	0.0478661
82%	0.052731
81%	0.0575959
80%	0.063208
79%	0.0695501
78%	0.0758922
77%	0.082643
76%	0.0901451
75%	0.0976472
74%	0.105976
73%	0.114683
72%	0.12389
71%	0.13388
70%	0.144347
69%	0.155568
68%	0.167476
67%	0.179834
66%	0.19325
65%	0.207197
64%	0.221782
63%	0.237264
62%	0.253565
61%	0.270635
60%	0.288534
59%	0.30731
58%	0.327011
57%	0.347675
56%	0.369334
55%	0.392004
54%	0.415694
53%	0.440432
52%	0.466404
51%	0.493477
50%	0.52171
49%	0.55126
48%	0.581945
47%	0.61393
46%	0.647138
45%	0.681616
44%	0.71753

43%	0.754927
42%	0.793842
41%	0.834306
40%	0.876324
39%	0.919908
38%	0.965176
37%	1.0121
36%	1.06069
35%	1.11112
34%	1.16336
33%	1.21752
32%	1.27363
31%	1.33168
30%	1.39178
29%	1.45406
28%	1.51853
27%	1.58516
26%	1.65387
25%	1.72484
24%	1.79842
23%	1.87489
22%	1.95428
21%	2.03659
20%	2.12188
19%	2.21028
18%	2.3018
17%	2.39646
16%	2.49406
15%	2.59464
14%	2.69878
13%	2.80723
12%	2.92012
11%	3.03748
10%	3.15972
9%	3.28744
8%	3.42045
7%	3.55802
6%	3.69944
5%	3.84498
4%	3.99602
3%	4.15372
2%	4.32043
1%	4.51272
0.9%	4.53563

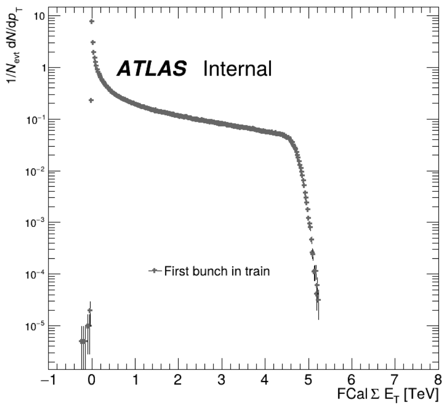


Figure 9: The FCal ΣE_T distribution of events corresponding to the first bunch crossing in a train.

0.8%	4.55983
0.7%	4.58593
0.6%	4.61415
0.5%	4.64539
0.4%	4.68082
0.3%	4.72314
0.2%	4.77714
0.1%	4.85819

Table 1: Centile cuts for FCalΣE_T, in 1% intervals.

4.1 Effects of Out-of-time pileup on the $FCal\Sigma E_T$ distribution

In order to assess the effect of Out-of-time pileup on the FCal ΣE_T distribution, a distribution free from Out-of-time pileup was constructed. This distribution was constructed by only taking events which contained the bunch crossing identification (BCID) corresponding to the first bunch in a train, and then applying the In-time cuts on it. The BCID information was taken from the ATLAS RunQuery online tool. This distribution is shown in Fig. 9. A parallel centrality analysis was performed on this distribution, and the migration in centrality bins was assessed. The effect of the out-of-time pileup can then be quantified by looking at the migration of events between the centrality bins. By taking the centile bins defined in Table 1 and taking the ratio between the number of events in each bin from both the first-in-train and nominal distributions, we can get a quantity which will correspond to the migration of events between the bins. This is shown in Table 2, in 1% wide bins from 80 to 60 % centrality.

1% intervals					
Centile cut	FCalΣE_T Total	FCalΣE_T First in Train	δ	$\delta/FCal_{total}$	$N^{\text{nominal}}_{\text{entries}}/N^{first}_{\text{entries}}$
80%	0.06373	0.06265	0.00108	0.016977	1.00614
79%	0.07008	0.06896	0.00112	0.0159765	1.0077
78%	0.07643	0.07527	0.00116	0.0151421	1.0077

77%	0.08326	0.08186	0.00141	0.0168857	1.00775
76%	0.09073	0.08928	0.00145	0.015993	1.00781
75%	0.09820	0.09670	0.00150	0.0152361	1.00781
74%	0.10660	0.10482	0.00178	0.0167014	1.00562
73%	0.11529	0.11348	0.00181	0.015682	1.00492
72%	0.12458	0.12247	0.00211	0.0169499	1.0047
71%	0.13459	0.13245	0.00214	0.0158916	1.00445
70%	0.14516	0.14272	0.00244	0.0167774	1.00488
69%	0.15640	0.15392	0.00248	0.0158344	1.0054
68%	0.16840	0.16563	0.00277	0.0164721	1.00581
67%	0.18083	0.17794	0.00289	0.0159973	1.00601
66%	0.19427	0.19114	0.00313	0.0161093	1.00614
65%	0.20830	0.20487	0.00343	0.0164865	1.00607
64%	0.22297	0.21928	0.00369	0.0165525	1.00602
63%	0.23846	0.23465	0.00381	0.0159656	1.006
62%	0.25484	0.25072	0.00412	0.0161729	1.00599
61%	0.27200	0.26756	0.00444	0.0163236	1.00601

Table 2: Centrality bin migration table due to out-of-time pileup in 1% intervals.

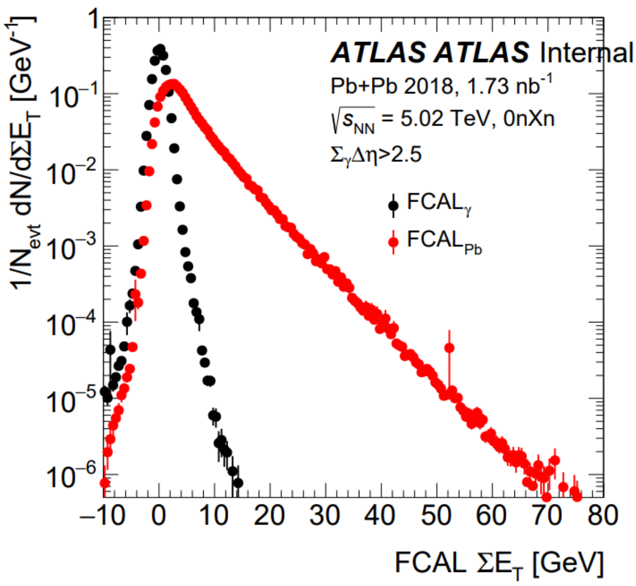


Figure 10: The FCal ΣE_T distribution for minimum bias Pb+Pb events at 5.02 TeV in 2018.

The working point chosen based on data from 2018 and Fig 10. is 30 GeV. However, a study of the amount of contamination from photonuclear and diffractive processes is still necessary. One potential way to do this is to look at the asymmetry in the FCal ΣE_T distribution. We define this asymmetry as:

$$A = \left(\mathrm{FCal}\Sigma E_T^A - \mathrm{FCal}\Sigma E_T^C\right) / \left(\mathrm{FCal}\Sigma E_T^A + \mathrm{FCal}\Sigma E_T^C\right) \tag{8}$$

The asymmetry was computed for all of the minimum bias Pb + Pb events in 2023. It is shown in Fig. 11, for 0.01 to 0.05 TeV. The projection of the asymmetry onto the FCal ΣE_T axis is shown in Fig. 12. The projection can be fit with a sum of three gaussian functions, contributing to the maxima at at -1, 0 and 1. Extending the individual gaussian fits to a larger range, we can see the intersection between the central gaussian (representing the signal) and the side gaussian, representing the contamination. The ratio of the integral of the backgoround gaussians to the total integral gives an estimate of the contamination. This is shown in Table 3. The intersection between the signal and background gaussians fluctuates around 0.9. In

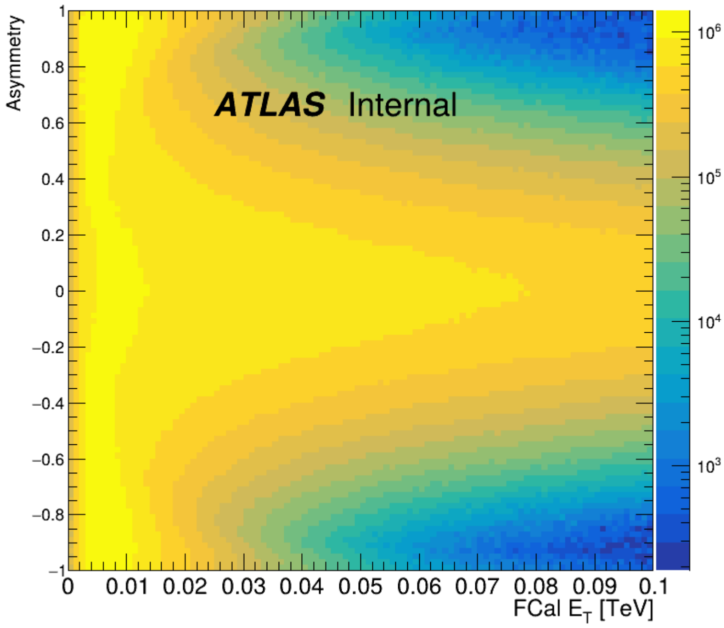


Figure 11: The asymmetry in the FCal ΣE_T distribution for minimum bias Pb+Pb events, shown up to 0.05 TeV.

order to reduce the contamination from the background, a kinematic cut of $A < 0.9$ is applied. The effect of this cut is non-negligible only in the 0-10% centrality bin, where it represents an overall removal of 3% of events. This is shown in Table 4. The events removed due to the kinematic cut are shown in Fig. 13.

10% intervals			
Centrality Bin	Signal Events	Events with |A|>0.9	Ratio
80% - 85%	9.9511e+08	3.00793e+07	0.0302271
70% - 80%	1.25182e+09	690258	0.000551403
60% - 70%	1.2291e+09	167370	0.000136173
50% - 60%	1.20686e+09	51502.6	4.2675e-05
40% - 50%	1.19028e+09	2128.81	1.7885e-06
30% - 40%	1.18417e+09	0	0
20% - 30%	1.22505e+09	0	0
10% - 20%	1.28375e+09	0	0
0% - 10%	1.24311e+09	0	0

Table 4: Proportion of events removed due to the Asymmetry cut in 10% centrality bins.

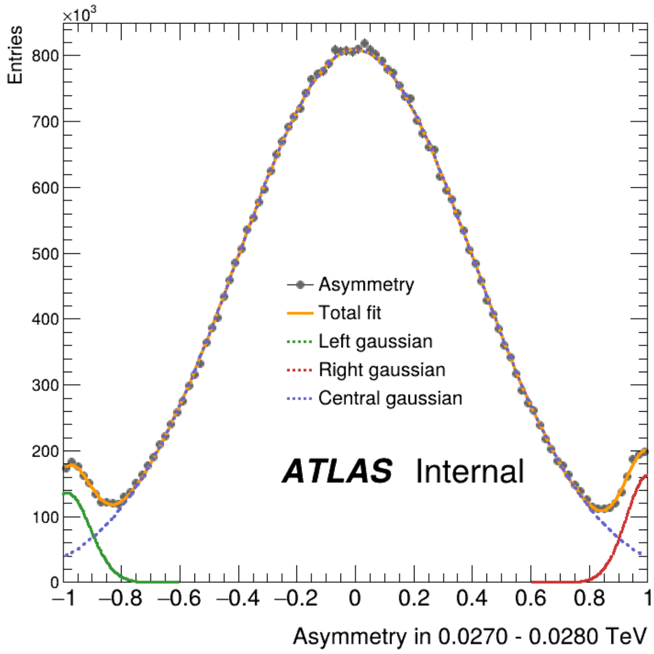


Figure 12: The projection of the asymmetry in the FCal ΣE_T distribution between 0.03 and 0.031 TeV. The projection is fit with a sum of three gaussian functions. The intersection between the central gaussian and the side gaussian is used to estimate the contamination. It is also used as a kinematic cut on the FCal ΣE_T distribution.

FCalΣE_T	Crossing < 0	Crossing > 0	contribution
0.025000 TeV	-0.888909	0.893993	0.039211
0.026000 TeV	-0.891965	0.896923	0.035156
0.027000 TeV	-0.895234	0.899036	0.031446
0.028000 TeV	-0.894808	0.901717	0.028929
0.029000 TeV	-0.896404	0.902380	0.026305
0.030000 TeV	-0.898972	0.902285	0.023689
0.031000 TeV	-0.902798	0.907236	0.020689
0.032000 TeV	-0.903520	0.911447	0.018742
0.033000 TeV	-0.908075	0.907915	0.016888
0.034000 TeV	-0.907720	0.916059	0.014539
0.035000 TeV	-0.910368	0.916794	0.013317
0.036000 TeV	-0.912967	0.918438	0.011461
0.037000 TeV	-0.916528	0.920813	0.009997
0.038000 TeV	-0.917558	0.924857	0.009036
0.039000 TeV	-0.917953	0.927993	0.007880
0.040000 TeV	-0.923000	0.924450	0.007082

Table 3: Crossing points and contamination estimation for values of ΣE_T between 0.025 and 0.040 TeV

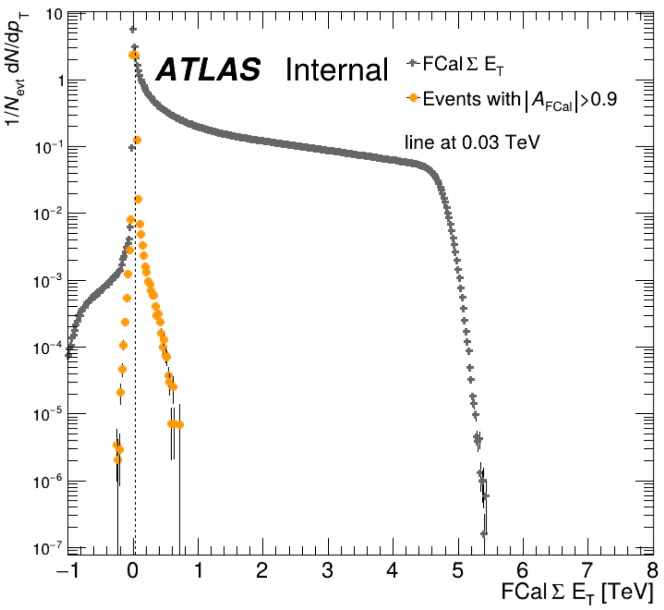


Figure 13: The FCal ΣE_T distribution for minimum bias Pb+Pb events after the kinematic cut on the asymmetry, along with the events rejected due to the cut.

4.2 Efficiency calculation

The efficiency is defined here as the proportion of Glauber-like cross section above the working point to the total Glauber-like cross section. The total Glauber-like cross section is estimated based on the integral of the simulated FCal ΣE_T distribution. The Glauber-like cross section above the working point is the integral of the Minimum Bias FCal ΣE_T distribution above the working point.

5 Systematic uncertainties

For this analysis, two systematic uncertainties were considered, those being uncertainties arising from the Glauber model and the calculation of the efficiencies. The uncertainties are added in quadrature, and the results are presented below.

5.1 Geometric uncertainties

To consider the possible uncertainties in the Glauber model, six alternative sets of Glauber events are generated, to account for the "up" and "down" variations of the input parameters. The parameters varied are:

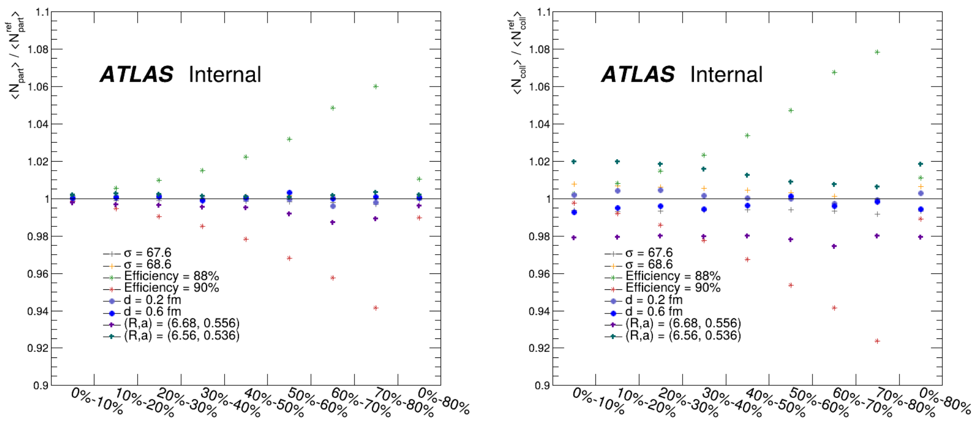
- Nucleon-nucleon cross section σ_{NN}: 67.1±0.5 mb
- Woods-Saxon parameters R, a: 6.62±0.06 fm, 0.546±0.01 fm

- Nucleon hard core d_N: 0.4± 0.2 fm

The fit to data was repeated using the alternative N_{elem} distributions, and the resulting geometric quantities were compared to the nominal values. The resulting uncertainties are shown in Fig. 14.

5.2 Efficiency uncertainties

The efficiency uncertainties are calculated by artifically varying the efficiency by ±1%. This is equivalent to multiplying the FCal ΣE_T interval by 90/89 or 88/89. The resulting uncertainties are shown in Fig. 14.



(a) Uncertainty in N_{part}

(b) Uncertainty in N_{coll}

Figure 14: Uncertainties in N_{part} and N_{coll} due to geometric and efficiency uncertainties.

6 Geometric quantities with systematic uncertainties

Presented here are the results for the geometric model and associated uncertainties. The results are shown in Tables 5, 6 and 7.

range	N_{coll}	δN_{coll}	$\delta N_{\text{coll}}/N_{\text{coll}}$
1% intervals for 0%-10%			
0% - 1%	1967.46	32.3496	0.0164423
1% - 2%	1861.36	35.8243	0.0192464
2% - 3%	1775.7	32.1326	0.0180958
3% - 4%	1688.85	33.2871	0.0197099
4% - 5%	1608.26	32.3618	0.0201222
5% - 6%	1531.18	30.4621	0.0198945
6% - 7%	1455.04	33.7295	0.0231812
7% - 8%	1379.16	36.0529	0.0261412
8% - 9%	1309.28	44.8938	0.0342889
9% - 10%	1245.7	49.3104	0.0395845
5% intervals for 0%-80%			
0% - 5%	1780.33	32.4318	0.0182168
5% - 10%	1384.07	38.5359	0.0278425
10% - 15%	1077.76	44.4333	0.0412273
15% - 20%	835.875	48.8595	0.0584531
20% - 25%	655.427	36.9025	0.056303
25% - 30%	503.278	23.3482	0.0463922
30% - 35%	378.95	22.8899	0.0604034
35% - 40%	287.665	17.4987	0.0608301
40% - 45%	213.647	8.61112	0.0403053
45% - 50%	147.571	9.46353	0.0641288
50% - 55%	101.213	8.08968	0.0799274
55% - 60%	68.2628	8.59272	0.125877
60% - 65%	44.6654	7.52659	0.168511
65% - 70%	29.2488	6.44222	0.220256
70% - 75%	19.3854	5.29311	0.273047
75% - 80%	11.6296	2.54688	0.219
10% intervals for 0%-80%			
0% - 10%	1582.2	34.8788	0.0220445
10% - 20%	956.819	46.5882	0.0486907
20% - 30%	579.352	30.0261	0.0518271
30% - 40%	333.308	20.1849	0.0605592
40% - 50%	180.609	7.48698	0.0414541
50% - 60%	84.7378	8.24506	0.0973009
60% - 70%	36.9571	6.98589	0.189027
70% - 80%	15.5075	3.91757	0.252625
20% intervals for 0%-100%			
0% - 20%	1269.51	39.9608	0.0314774
20% - 40%	456.33	25.0296	0.0548498
40% - 60%	132.673	7.12257	0.0536849
60% - 80%	26.2323	5.45195	0.207833
80% - 100%	3.8142	1.03312	0.270861

0.1% intervals for 0%-1%			
0% - 0.1%	2066.04	42.269	0.0204589
0.1% - 0.2%	2021.61	36.7532	0.0181801
0.2% - 0.3%	1997.07	32.6961	0.016372
0.3% - 0.4%	1979.34	32.1639	0.0162498
0.4% - 0.5%	1963.87	33.1178	0.0168636
0.5% - 0.6%	1950.79	34.5222	0.0176965
0.6% - 0.7%	1939.87	35.212	0.0181517
0.7% - 0.8%	1928.39	37.144	0.0192617
0.8% - 0.9%	1918.44	36.7802	0.019172
0.9% - 1%	1909.18	36.968	0.0193633

Table 5: N_{coll} values for different centrality intervals.

range	N_{part}	δN_{part}	$\delta N_{\text{part}}/N_{\text{part}}$
1% intervals for 0%-10%			
0% - 1%	400.273	1.41676	0.00353948
1% - 2%	391.976	3.08755	0.00787689
2% - 3%	382.951	2.29978	0.00600542
3% - 4%	372.698	3.12211	0.00837705
4% - 5%	362.624	3.22863	0.00890354
5% - 6%	352.227	1.94414	0.00551957
6% - 7%	341.288	2.13603	0.00625871
7% - 8%	330.132	2.65284	0.00803568
8% - 9%	319.081	3.88846	0.0121864
9% - 10%	308.266	4.99506	0.0162037
5% intervals for 0%-80%			
0% - 5%	382.104	2.62492	0.00686965
5% - 10%	330.199	3.11876	0.00944511
10% - 15%	279.651	7.67047	0.0274287
15% - 20%	237.916	8.63309	0.0362863
20% - 25%	201.35	5.50535	0.0273422
25% - 30%	169.045	6.06365	0.0358701
30% - 35%	140.337	5.48666	0.0390963
35% - 40%	117.929	4.85194	0.0411431
40% - 45%	96.5638	1.78583	0.0184938
45% - 50%	76.1643	2.33333	0.0306354
50% - 55%	58.9451	2.26051	0.0383493
55% - 60%	45.0525	2.91426	0.064686
60% - 65%	33.5643	3.36382	0.10022
65% - 70%	24.6801	3.11532	0.126228
70% - 75%	18.09	2.85898	0.158041
75% - 80%	12.1998	1.87698	0.153853
10% intervals for 0%-80%			
0% - 10%	356.152	2.85725	0.00802257
10% - 20%	258.783	8.15292	0.0315048
20% - 30%	185.197	5.78552	0.0312398
30% - 40%	129.133	5.16852	0.0400248
40% - 50%	86.364	1.99419	0.0230905
50% - 60%	51.9988	2.4912	0.0479088
60% - 70%	29.1222	3.2397	0.111245
70% - 80%	15.1449	2.36797	0.156354
20% intervals for 0%-100%			
0% - 20%	307.467	5.52544	0.0179708
20% - 40%	157.165	5.47219	0.0348181
40% - 60%	69.1814	2.04675	0.0295852
60% - 80%	22.1336	2.80359	0.126667
80% - 100%	4.98987	0.849331	0.170211

0.1% intervals for 0%-1%			
0% - 0.1%	405.273	0.26162	0.000645539
0.1% - 0.2%	403.582	0.23871	0.000591478
0.2% - 0.3%	402.384	0.381894	0.00094908
0.3% - 0.4%	401.388	0.788252	0.00196382
0.4% - 0.5%	400.425	1.22002	0.0030468
0.5% - 0.6%	399.544	1.62621	0.00407018
0.6% - 0.7%	398.764	2.06334	0.00517434
0.7% - 0.8%	397.898	2.29009	0.00575548
0.8% - 0.9%	397.114	2.56022	0.00644706
0.9% - 1%	396.355	2.79033	0.00703997

Table 6: N_{part} values for different centrality intervals.

range	T_{AA}	δT_{AA}	$\delta T_{\mathrm{AA}}/T_{\mathrm{AA}}$
1% intervals for 0%-10%			
0% - 1%	28.8908	0.475031	0.0164423
1% - 2%	27.3327	0.526055	0.0192464
2% - 3%	26.0749	0.471844	0.0180958
3% - 4%	24.7995	0.488797	0.0197099
4% - 5%	23.6162	0.47521	0.0201222
5% - 6%	22.4843	0.447314	0.0198945
6% - 7%	21.3662	0.495294	0.0231812
7% - 8%	20.252	0.529411	0.0261412
8% - 9%	19.2259	0.659234	0.0342889
9% - 10%	18.2922	0.724088	0.0395845
5% intervals for 0%-80%			
0% - 5%	326.1428	0.476237	0.0182168
5% - 10%	20.3241	0.565873	0.0278425
10% - 15%	15.8262	0.652471	0.0412273
15% - 20%	12.2742	0.717466	0.0584531
20% - 25%	9.62447	0.541887	0.056303
25% - 30%	7.39028	0.342852	0.0463922
30% - 35%	5.56462	0.336122	0.0604034
35% - 40%	4.22415	0.256956	0.0608301
40% - 45%	3.13726	0.126448	0.0403053
45% - 50%	2.16697	0.138965	0.0641288
50% - 55%	1.48624	0.118791	0.0799274
55% - 60%	1.00239	0.126178	0.125877
60% - 65%	0.65588	0.110523	0.168511
65% - 70%	0.429498	0.0945994	0.220256
70% - 75%	0.28466	0.0777256	0.273047
75% - 80%	0.170772	0.0373992	0.219
10% intervals for 0%-80%			
0% - 10%	23.2335	0.512171	0.0220445
10% - 20%	14.0502	0.684114	0.0486907
20% - 30%	8.50738	0.440913	0.0518271
30% - 40%	4.89439	0.2964	0.0605592
40% - 50%	2.65212	0.109941	0.0414541
50% - 60%	1.24431	0.121073	0.0973009
60% - 70%	0.542689	0.102583	0.189027
70% - 80%	0.227716	0.0575267	0.252625
20% intervals for 0%-100%			
0% - 20%	18.6418	0.586796	0.0314774
20% - 40%	6.70088	0.367542	0.0548498
40% - 60%	1.94822	0.10459	0.0536849
60% - 80%	0.385203	0.0800579	0.207833
80% - 100%	0.0560088	0.0151706	0.270861

0.1% intervals for 0%-1%			
0% - 0.1%	30.3384	0.620691	0.0204589
0.1% - 0.2%	29.686	0.539694	0.0181801
0.2% - 0.3%	29.3256	0.48012	0.016372
0.3% - 0.4%	29.0653	0.472304	0.0162498
0.4% - 0.5%	28.838	0.486311	0.0168635
0.5% - 0.6%	28.646	0.506933	0.0176965
0.6% - 0.7%	28.4856	0.517063	0.0181517
0.7% - 0.8%	28.317	0.545433	0.0192617
0.8% - 0.9%	28.1709	0.540091	0.019172
0.9% - 1%	28.0349	0.542849	0.0193633

Table 7: T_{AA} values for different centrality intervals.

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